

Case Study #4:

Research Project Management Office (RMPO)

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about research projects, grants, donors, and the faculty, staff, and students involved.

OBJECTIVE

1. Maintain Data with CRUD method

Administrators can retrieve, review, and update information within the Research Project Management Office (RMPO) system by using standard CRUD (Create, Read, Update, Delete) operations to ensure accurate and up-to-date database management.

2. Analyse and report statistics for management

Administrators can utilize the data stored in the system to perform statistical analysis and generate reports. These insights can serve as reliable references to support decision-making and strategic management.

3. Tracking, planning and forecasting

Administrators can track the progress of each research project and leverage project data for planning purposes. The system also allows forecasting of project timelines, required resources, or budgets to improve efficiency and project outcomes.

4. Research performance of each faculty

Administrators can evaluate the research performance of each faculty or department, enabling the assessment of capabilities, achievements, and overall contributions to institutional research success.

Kookle University
Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

Keywords: Ontology, Plant Disease, Intelligent System

Project Category: ☐ Research and Development
☐ Training/Capacity Building
☐ Conference/Workshop

Research Areas of the Project: Semantic Web, Ontology, Knowledge Representation

Project Duration: 12 months

Contract Period: 1 Jan 2019 – 31 Dec 2019

Budget: 1,000,000 THB

Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:

1. Name: Mark Zuckerberg Email: mark@kmail.com
Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
Department ICT Department Area of Expertise: Knowledge Representation

Project Member:

1. Name: Tim Berners Lee Email: tim@kmail.com
Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: _____ Email: _____
Department _____ Area of Expertise: _____

Donor/Sponsor Name: Thailand Research Fund

Donor Type: ☐ Research Organization
☐ International Organization
☐ Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:

1. Name: Nina Billa Email: nina@kmail.com
Department: ICT
Responsibility: Programming
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
Department: Business Management
Responsibility: Project Coordinator
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Jan 2019 End date: 31 Dec 2019

Based on the form, the requirements for the Kookle University Research Project Management Office (RPMO) system are as follows:

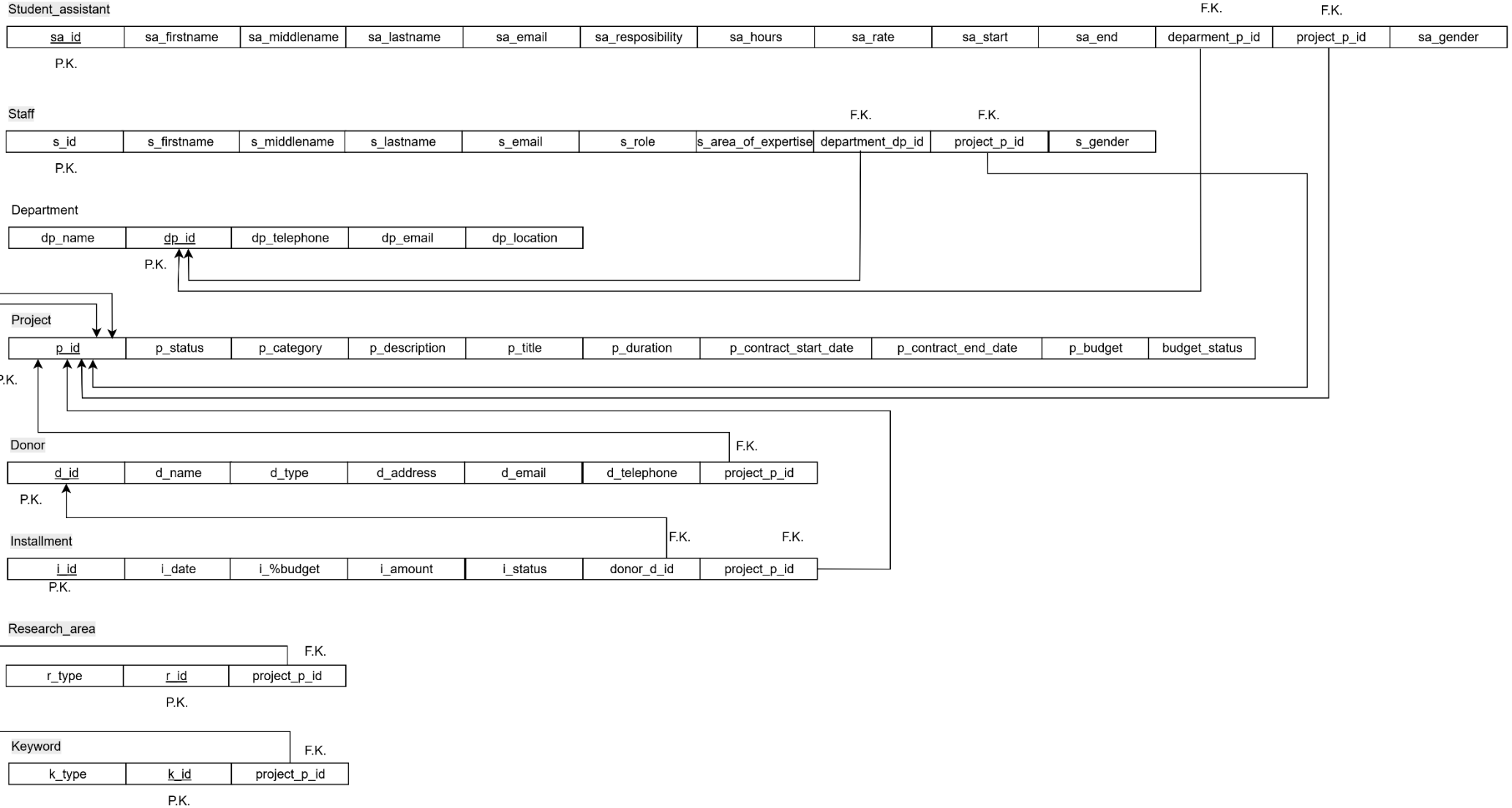
Data Maintenance: The system must maintain data about research projects, grants received, donors, and the faculty, staff, and students working on the projects.

Data Entry and Updates: It should support the entry and updating of project-related data after a project form is submitted.

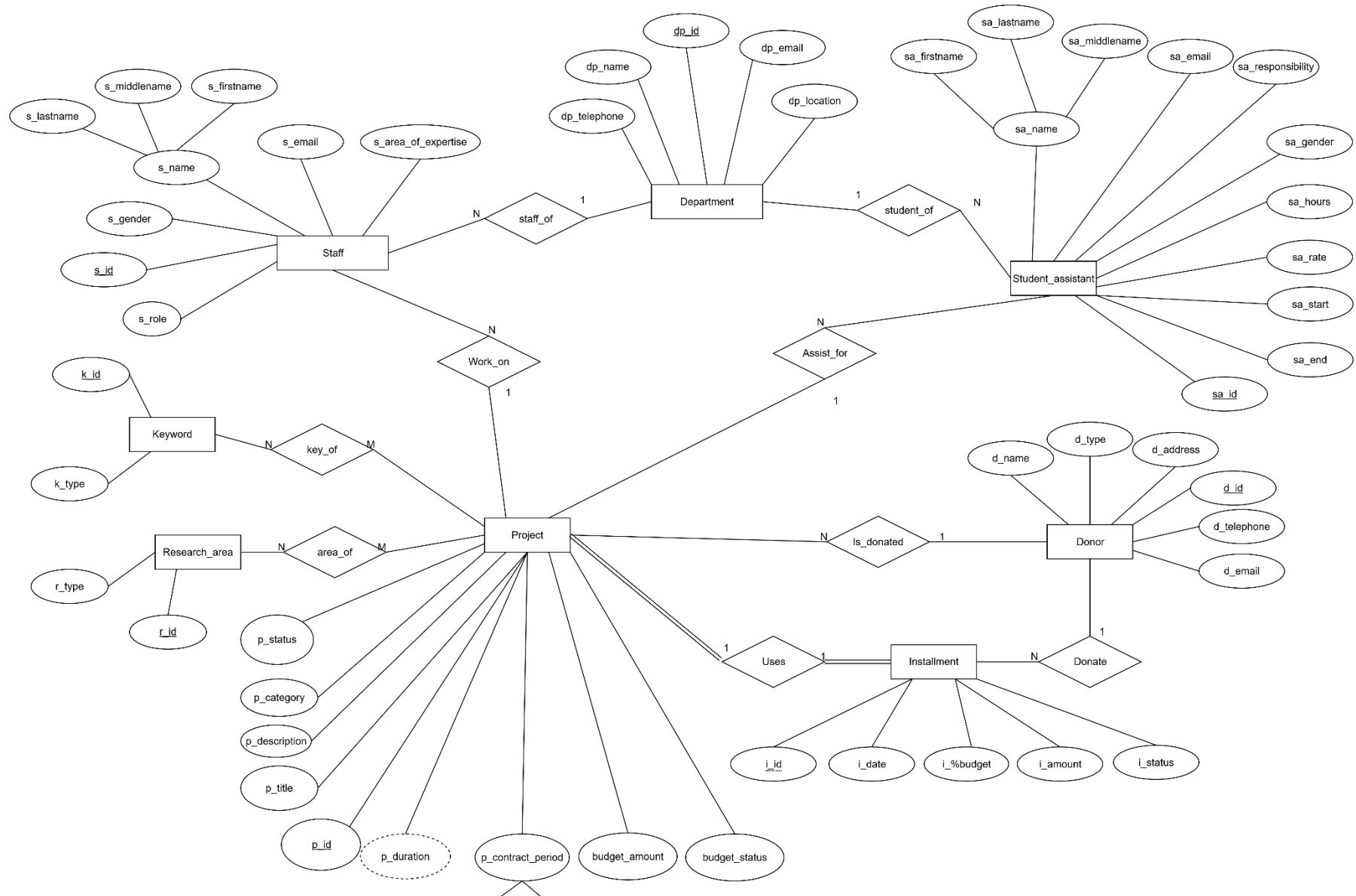
Querying: The system needs to allow for querying data from project spreadsheets. This includes queries to coordinate with the project investigator.

Reporting: The system should be able to generate useful reports or statistics for university management. Examples of reports include total grants received for each active research area, statistics about sponsors/donors, and the research performance of each faculty/staff member.

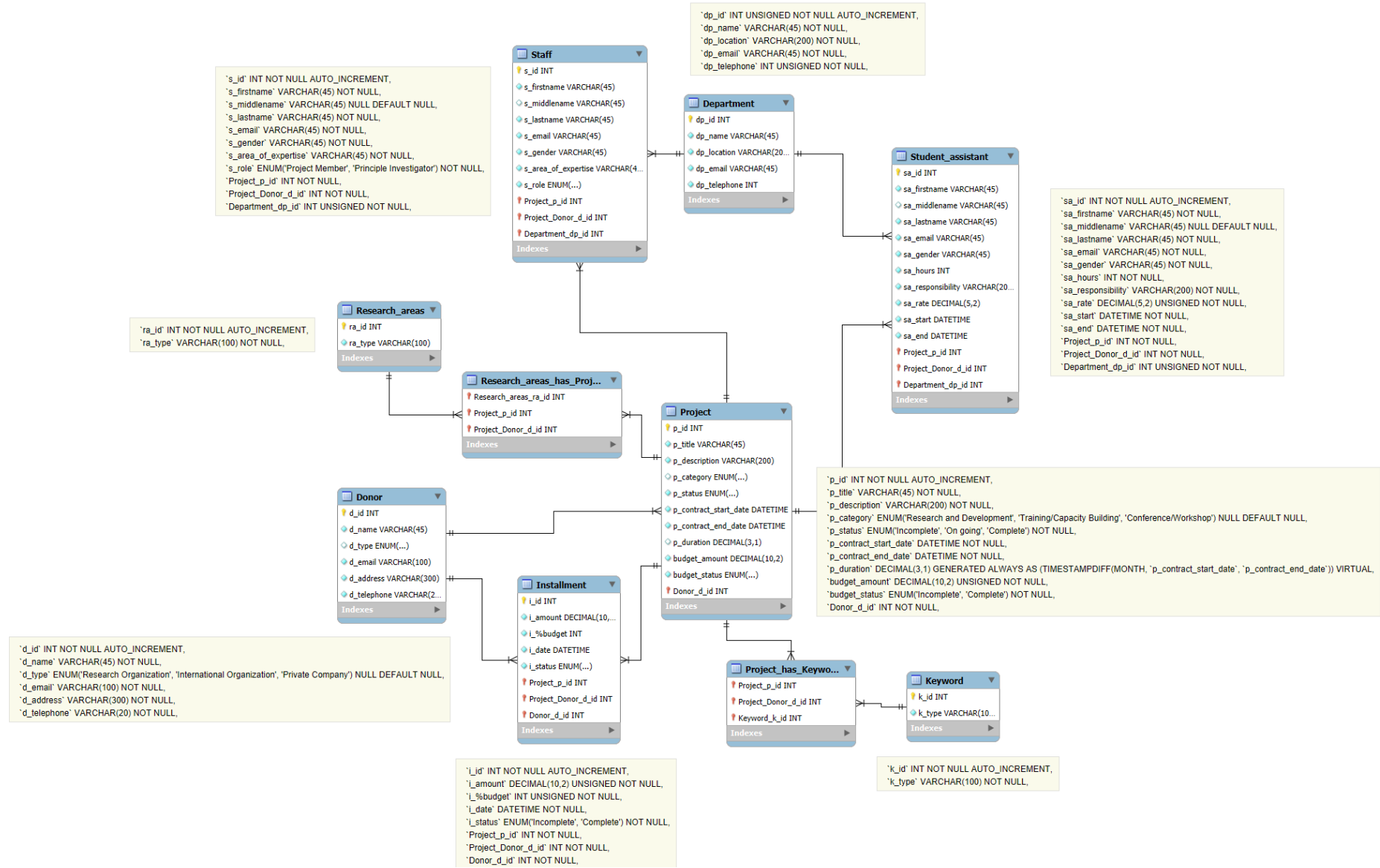
Relational database schema



Conceptual diagram



ER- diagram



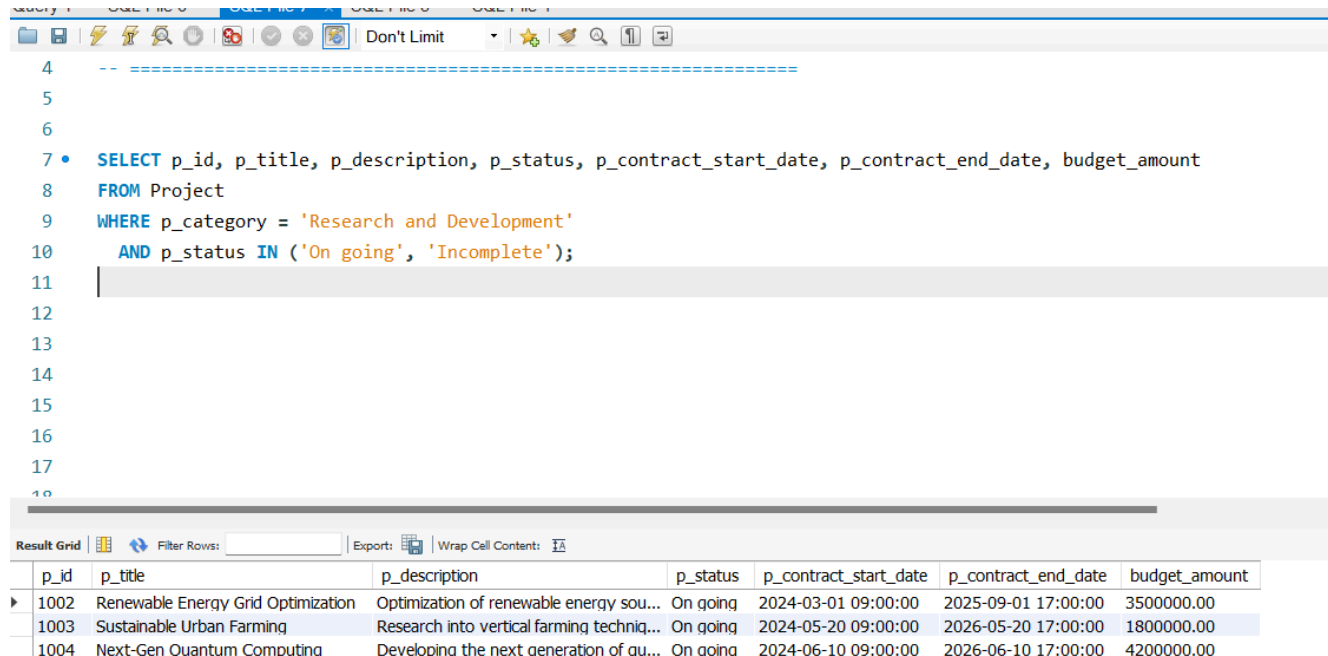
Work Distribution and Contribution

QUERY

1.Find all active projects in "Research and Development" category

SQL Command:

```
SELECT p_id, p_title, p_description, p_status, p_contract_start_date, p_contract_end_date, budget_amount
FROM Project
WHERE p_category = 'Research and Development'
AND p_status IN ('On going', 'Incomplete');
```



The screenshot shows a SQL query editor with a toolbar at the top. The query is entered in the editor area. Below the editor is a 'Result Grid' showing the output of the query. The grid has columns for p_id, p_title, p_description, p_status, p_contract_start_date, p_contract_end_date, and budget_amount. Three rows of data are displayed.

p_id	p_title	p_description	p_status	p_contract_start_date	p_contract_end_date	budget_amount
1002	Renewable Energy Grid Optimization	Optimization of renewable energy sou...	On going	2024-03-01 09:00:00	2025-09-01 17:00:00	3500000.00
1003	Sustainable Urban Farming	Research into vertical farming techniq...	On going	2024-05-20 09:00:00	2026-05-20 17:00:00	1800000.00
1004	Next-Gen Quantum Computing	Developing the next generation of qu...	On going	2024-06-10 09:00:00	2026-06-10 17:00:00	4200000.00

2. List all projects funded by donors of type "Research Organization"

SQL Command:

```
SELECT p.p_id, p.p_title, p.p_category, p.budget_amount,  
       d.d_name AS donor_name, d.d_email, d.d_telephone  
FROM Project p  
JOIN Donor d ON p.Donor_d_id = d.d_id  
WHERE d.d_type = 'Research Organization'  
ORDER BY p.p_title;
```

```
20 • SELECT p.p_id, p.p_title, p.p_category, p.budget_amount,  
21       d.d_name AS donor_name, d.d_email, d.d_telephone  
22 FROM Project p  
23 JOIN Donor d ON p.Donor_d_id = d.d_id  
24 WHERE d.d_type = 'Research Organization'  
25 ORDER BY p.p_title;
```

p_id	p_title	p_category	budget_amount	donor_name	d_email	d_telephone
1001	AI-Powered Drug Discovery	Research and Development	2500000.00	FutureTech Foundation	contact@futuretech.org	+1-555-0101
1007	Early Childhood Education Models	Training/Capacity Building	1200000.00	Innovative Minds NGO	contact@innovativeminds.org	+33-1-555-0107
1009	Historical Preservation Techniques	Conference/Workshop	1100000.00	Foundation for the Arts	arts@foundation.org	+39-06-555-0109
1005	Marine Biodiversity Study	Training/Capacity Building	1500000.00	Environmental Conservation Trust	info@ect.org	+44-20-7555-0105
1011	Microbial Fuel Cells	Conference/Workshop	2800000.00	Green Energy Alliance	contact@gea.com	+61-2-555-0111

3. Calculate total budget amount by research area for current year

SQL Command:

```
SELECT ra.ra_type AS research_area,  
       COUNT(p.p_id) AS project_count,  
       SUM(p.budget_amount) AS total_budget,  
       AVG(p.budget_amount) AS average_budget  
FROM Research_areas ra  
JOIN Research_areas_has_Project rahp ON ra.ra_id = rahp.Research_areas_ra_id  
JOIN Project p ON rahp.Project_p_id = p.p_id AND rahp.Project_Donor_d_id = p.Donor_d_id  
WHERE YEAR(p.p_contract_start_date) = YEAR(CURDATE())  
GROUP BY ra.ra_type  
ORDER BY total_budget DESC;
```

```
31 • SELECT ra.ra_type AS research_area,  
32        COUNT(p.p_id) AS project_count,  
33        SUM(p.budget_amount) AS total_budget,  
34        AVG(p.budget_amount) AS average_budget  
35 FROM Research_areas ra  
36 JOIN Research_areas_has_Project rahp ON ra.ra_id = rahp.Research_areas_ra_id  
37 JOIN Project p ON rahp.Project_p_id = p.p_id AND rahp.Project_Donor_d_id = p.Donor_d_id  
38 WHERE YEAR(p.p_contract_start_date) = YEAR(CURDATE())  
39 GROUP BY ra.ra_type  
40 ORDER BY total_budget DESC;
```

41

42

4. Show project details with donor information and installment schedule

SQL Command:

```
SELECT p.p_id, p.p_title, p.p_status, p.budget_amount,
       d.d_name AS donor_name, d.d_type, d.d_email, d.d_address,
       i.i_id AS installment_id, i.i_amount, i.i_date, i.i_status,
       CONCAT(i.`i_%budget`, '%') AS budget_percentage
FROM Project p
JOIN Donor d ON p.Donor_d_id = d.d_id
LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
ORDER BY p.p_id, i.i_date;
```

```
46 • SELECT p.p_id, p.p_title, p.p_status, p.budget_amount,
47       d.d_name AS donor_name, d.d_type, d.d_email, d.d_address,
48       i.i_id AS installment_id, i.i_amount, i.i_date, i.i_status,
49       CONCAT(i.`i_%budget`, '%') AS budget_percentage
50 FROM Project p
51 JOIN Donor d ON p.Donor_d_id = d.d_id
52 LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
53 ORDER BY p.p_id, i.i_date;
```

Result Grid									
Filter Rows: Export: Wrap Cell Content:									
	p_id	p_title	p_status	budget_amount	donor_name	d_type	d_email	d_address	installment_id
▶	1001	AI-Powered Drug Discovery	Complete	2500000.00	FutureTech Foundation	Research Organization	contact@futuretech.org	123 Tech Avenue, New York, NY	2001
	1001	AI-Powered Drug Discovery	Complete	2500000.00	FutureTech Foundation	Research Organization	contact@futuretech.org	123 Tech Avenue, New York, NY	2002

5. Identify projects ending within next 3 months

SQL Command:

```
SELECT p_id, p_title, p_contract_end_date, p_status,  
       DATEDIFF(p_contract_end_date, CURDATE()) AS days_remaining  
FROM Project  
WHERE p_contract_end_date BETWEEN CURDATE() AND DATE_ADD(CURDATE(), INTERVAL 3 MONTH)  
     AND p_status != 'Complete'  
ORDER BY p_contract_end_date ASC;
```

```
59 • SELECT p_id, p_title, p_contract_end_date, p_status,  
60       DATEDIFF(p_contract_end_date, CURDATE()) AS days_remaining  
61 FROM Project  
62 WHERE p_contract_end_date BETWEEN CURDATE() AND DATE_ADD(CURDATE(), INTERVAL 3 MONTH)  
63       AND p_status != 'Complete'  
64 ORDER BY p_contract_end_date ASC;  
65  
66  
67  
68  
69
```

6. Calculate average project duration by category

SQL Command:

```
SELECT p_category,  
       COUNT(*) AS project_count,  
       AVG(p_duration) AS average_duration_months,  
       MIN(p_duration) AS min_duration,  
       MAX(p_duration) AS max_duration  
FROM Project  
GROUP BY p_category  
ORDER BY average_duration_months DESC;
```

```
70 • SELECT p_category,  
71         COUNT(*) AS project_count,  
72         AVG(p_duration) AS average_duration_months,  
73         MIN(p_duration) AS min_duration,  
74         MAX(p_duration) AS max_duration  
75 FROM Project  
76 GROUP BY p_category  
77 ORDER BY average_duration_months DESC;  
78  
79  
80  
81
```

7. List all international organization donors and their contact details

SQL Command:

```
SELECT d_id, d_name, d_email, d_address, d_telephone  
FROM Donor  
WHERE d_type = 'International Organization'  
ORDER BY d_name;
```

```
95 • SELECT d_id, d_name, d_email, d_address, d_telephone  
96 FROM Donor  
97 WHERE d_type = 'International Organization'  
98 ORDER BY d_name;  
99  
100  
101  
102  
103  
104  
105
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	d_id	d_name	d_email	d_address	d_telephone
▶	103	Global Health Corp	info@globalhealth.com	789 Global Road, Geneva, Switzerland	+41-22-555-0103

8. Calculate total funding received from each donor type by year

SQL Command:

```
SELECT d.d_type AS donor_type, bYEAR(p.p_contract_start_date) AS project_year, COUNT(p.p_id) AS projects_funded,  
       SUM(p.budget_amount) AS total_budget_allocated, SUM(COALESCE(inst_total.received_amount, 0)) AS  
total_amount_received  
FROM Donor d  
JOIN Project p ON d.d_id = p.Donor_d_id  
LEFT JOIN (  
    SELECT Project_p_id, Project_Donor_d_id, SUM(CASE WHEN i_status = 'Complete' THEN i_amount ELSE 0 END) AS  
received_amount  
    FROM Installment  
    GROUP BY Project_p_id, Project_Donor_d_id  
) inst_total ON p.p_id = inst_total.Project_p_id AND p.Donor_d_id = inst_total.Project_Donor_d_id  
GROUP BY d.d_type, YEAR(p.p_contract_start_date)  
ORDER BY project_year DESC, total_budget_allocated DESC;
```

```

104 • SELECT d.d_type AS donor_type,
105         YEAR(p.p_contract_start_date) AS project_year,
106         COUNT(p.p_id) AS projects_funded,
107         SUM(p.budget_amount) AS total_budget_allocated,
108         SUM(COALESCE(inst_total.received_amount, 0)) AS total_amount_received
109 FROM Donor d
110 JOIN Project p ON d.d_id = p.Donor_d_id
111 LEFT JOIN (
112     SELECT Project_p_id, Project_Donor_d_id,
113            SUM(CASE WHEN i_status = 'Complete' THEN i_amount ELSE 0 END) AS received_amount
114     FROM Installment
115     GROUP BY Project_p_id, Project_Donor_d_id
116 ) inst_total ON p.p_id = inst_total.Project_p_id AND p.Donor_d_id = inst_total.Project_Donor_d_id
117 GROUP BY d.d_type, YEAR(p.p_contract_start_date)
118 ORDER BY project_year DESC, total_budget_allocated DESC;
119

```

donor_type	project_year	projects_funded	total_budget_allocated	total_amount_received
International Organization	2025	1	3000000.00	3000000.00
Research Organization	2025	1	2800000.00	2800000.00
International Organization	2024	4	10600000.00	11100000.00
Private Company	2024	2	9000000.00	9000000.00
Research Organization	2024	4	6300000.00	6300000.00

9. Identify top 5 research

SQL Command:

```

SELECT ra.ra_type AS research_area, COUNT(p.p_id) AS total_projects, SUM(p.budget_amount) AS total_funding_allocated,
       SUM(COALESCE(inst_received.amount_received, 0)) AS total_funding_received,
       ROUND((SUM(COALESCE(inst_received.amount_received, 0)) / SUM(p.budget_amount)) * 100, 2) AS funding_efficiency_percent
FROM Research_areas ra
JOIN Research_areas_has_Project rahp ON ra.ra_id = rahp.Research_areas_ra_id
JOIN Project p ON rahp.Project_p_id = p.p_id AND rahp.Project_Donor_d_id = p.Donor_d_id
LEFT JOIN (
    SELECT Project_p_id, Project_Donor_d_id, SUM(CASE WHEN i_status = 'Complete' THEN i_amount ELSE 0 END) AS
amount_received
    FROM Installment
    GROUP BY Project_p_id, Project_Donor_d_id
) inst_received ON p.p_id = inst_received.Project_p_id AND p.Donor_d_id = inst_received.Project_Donor_d_id

```

GROUP BY ra.ra_type

ORDER BY total_funding_received DESC

LIMIT 5;

```
130     SELECT Project_p_id, Project_Donor_d_id,  
131            SUM(CASE WHEN i_status = 'Complete' THEN i_amount ELSE 0 END) AS amount_received  
132     FROM Installment  
133     GROUP BY Project_p_id, Project_Donor_d_id  
134 ) inst_received ON p.p_id = inst_received.Project_p_id AND p.Donor_d_id = inst_received.Project_Donor_d_id  
135 GROUP BY ra.ra_type  
136 ORDER BY total_funding_received DESC  
137 LIMIT 5;
```

research_area	total_projects	total_funding_allocated	total_funding_received	funding_efficiency_percent
Computer Science	1	4800000.00	4800000.00	100.00
Quantum Physics	1	4200000.00	4200000.00	100.00
Electrical Engineering	1	3500000.00	4000000.00	114.29
Genetics	1	3200000.00	3200000.00	100.00
Urban Planning	1	3000000.00	3000000.00	100.00

10. Show project timeline and payment schedule

SQL Command:

SELECT p.p_id, p.p_title, p.p_contract_start_date, p.p_contract_end_date,

p.p_duration AS duration_months,

i.i_date AS payment_date, i.i_amount, i.i_status,

DATEDIFF(i.i_date, p.p_contract_start_date) AS days_from_start

FROM Project p

LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id

ORDER BY p.p_id, i.i_date;

```
143 • SELECT p.p_id, p.p_title, p.p_contract_start_date, p.p_contract_end_date,  
144        p.p_duration AS duration_months,  
145        i.i_date AS payment_date, i.i_amount, i.i_status,  
146        DATEDIFF(i.i_date, p.p_contract_start_date) AS days_from_start  
147 FROM Project p  
148 LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id  
149 ORDER BY p.p_id, i.i_date;  
150
```


11. List projects with their completion percentage based on installments received

SQL Command:

```
SELECT p.p_id, p.p_title, p.budget_amount,  
       COALESCE(SUM(CASE WHEN i.i_status = 'Complete' THEN i.i_amount ELSE 0 END), 0) AS amount_received,  
       ROUND((COALESCE(SUM(CASE WHEN i.i_status = 'Complete' THEN i.i_amount ELSE 0 END), 0) / p.budget_amount) *  
100, 2) AS completion_percentage,  
       p.p_status  
FROM Project p  
LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id  
GROUP BY p.p_id, p.p_title, p.budget_amount, p.p_status  
ORDER BY completion_percentage DESC;
```

```

157 • SELECT p.p_id, p.p_title, p.budget_amount,
158         COALESCE(SUM(CASE WHEN i.i_status = 'Complete' THEN i.i_amount ELSE 0 END), 0) AS amount_received,
159         ROUND((COALESCE(SUM(CASE WHEN i.i_status = 'Complete' THEN i.i_amount ELSE 0 END), 0) / p.budget_amount) * 100, 2) AS completion_percentage,
160         p.p_status
161 FROM Project p
162 LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
163 GROUP BY p.p_id, p.p_title, p.budget_amount, p.p_status
164 ORDER BY completion_percentage DESC;

```

Result Grid						
Filter Rows: Export: Wrap Cell Content: ☐						
p_id	p_title	budget_amount	amount_received	completion_percentage	p_status	
1002	Renewable Energy Grid Optimization	3500000.00	4000000.00	114.29	On going	
1001	AI-Powered Drug Discovery	2500000.00	2500000.00	100.00	Complete	
1003	Sustainable Urban Farming	1800000.00	1800000.00	100.00	On going	
1004	Next-Gen Quantum Computing	4200000.00	4200000.00	100.00	On going	
1005	Marine Biodiversity Study	1500000.00	1500000.00	100.00	Complete	
1006	Advanced Robotics for Manufacturing	2100000.00	2100000.00	100.00	On going	
1007	Early Childhood Education Models	1200000.00	1200000.00	100.00	On going	

12. Update project status when contract end date is reached

SQL Command:

UPDATE Project

SET p_status = 'Complete'

WHERE p_contract_end_date <= CURDATE()

AND p_status = 'On going'

AND budget_status = 'Complete';

```

185 • UPDATE Project
186 SET p_status = 'Complete'
187 WHERE p_contract_end_date <= CURDATE()
188 AND p_status = 'On going'
189 AND budget_status = 'Complete';

```

13. Insert new installment record for approved project payments

SQL Command:

```
UPDATE Project
SET p_status = 'Complete'
WHERE p_contract_end_date <= CURDATE()
      AND p_status = 'On going'
      AND budget_status = 'Complete';
```

```
202 • UPDATE Project
203 SET p_status = 'Complete'
204 WHERE p_contract_end_date <= CURDATE()
205     AND p_status = 'On going'
206     AND budget_status = 'Complete';
207
208
```

14. Delete projects ID = 1001 and associated records

SQL Command:

```
DELETE FROM Installment WHERE Project_p_id = 1001;
```

```
DELETE FROM Student_assistant WHERE Project_p_id = 1001;
```

```
DELETE FROM Staff WHERE Project_p_id = 1001;
```

```
DELETE FROM Project WHERE p_id = 1001;
```

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search. Below the toolbar, four SQL statements are listed, each preceded by a blue bullet point:

- 1 • `DELETE FROM Installment WHERE Project_p_id = 1001;`
- 2 • `DELETE FROM Student_assistant WHERE Project_p_id = 1001;`
- 3 • `DELETE FROM Staff WHERE Project_p_id = 1001;`
- 4 • `DELETE FROM Project WHERE p_id = 1001;`

Below the statements is an "Output" section with a dropdown menu set to "Action Output". It displays a table with the execution results:

#	Time	Action	Message
✓ 1	19:14:04	DELETE FROM Installment WHERE Project_p_id = 1001	4 row(s) affected
✓ 2	19:14:04	DELETE FROM Student_assistant WHERE Project_p_id = 1001	2 row(s) affected
✓ 3	19:14:04	DELETE FROM Staff WHERE Project_p_id = 1001	4 row(s) affected
✓ 4	19:14:04	DELETE FROM Project WHERE p_id = 1001	1 row(s) affected

15. Calculate monthly burn rate for active projects

SQL Command:

```
SELECT p.p_id, p.p_title, p.budget_amount, p.p_duration, ROUND(p.budget_amount / p.p_duration, 2) AS
monthly_budget_allocation,
    COALESCE(staff_cost.monthly_staff_cost, 0) AS monthly_staff_cost,
    COALESCE(student_cost.monthly_student_cost, 0) AS monthly_student_cost,
    ROUND((COALESCE(staff_cost.monthly_staff_cost, 0) + COALESCE(student_cost.monthly_student_cost, 0)), 2) AS
total_monthly_burn_rate
FROM Project p
```

```

LEFT JOIN (
    SELECT Project_p_id, Project_Donor_d_id, COUNT(*) * 5000 / p_duration AS monthly_staff_cost
    FROM Staff s
    JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
    GROUP BY Project_p_id, Project_Donor_d_id, p_duration
) staff_cost ON p.p_id = staff_cost.Project_p_id AND p.Donor_d_id = staff_cost.Project_Donor_d_id
LEFT JOIN (
    SELECT Project_p_id, Project_Donor_d_id,
        SUM(sa_hours * sa_rate * 4.33) AS monthly_student_cost
    FROM Student_assistant
    GROUP BY Project_p_id, Project_Donor_d_id
) student_cost ON p.p_id = student_cost.Project_p_id AND p.Donor_d_id = student_cost.Project_Donor_d_id
WHERE p.p_status = 'On going'
ORDER BY total_monthly_burn_rate DESC;

```

```

246     GROUP BY Project_p_id, Project_Donor_d_id
247 ) student_cost ON p.p_id = student_cost.Project_p_id AND p.Donor_d_id = student_cost.Project_Donor_d_id
248 WHERE p.p_status = 'On going'
249 ORDER BY total_monthly_burn_rate DESC;
250

```

Result Grid Filter Rows: Export: Wrap Cell Content:								
	p_id	p_title	budget_amount	p_duration	monthly_budget_allocation	monthly_staff_cost	monthly_student_cost	total_monthly_burn_rate
▶	1008	Cybersecurity Protocol Development	4800000.00	12.0	400000.00	1666.6667	3459.6700	5126.34
	1010	Genomic Sequencing of Rare Diseases	3200000.00	24.0	133333.33	833.3333	3897.0000	4730.33
	1012	Smart City Traffic Management	3000000.00	24.0	125000.00	833.3333	3117.6000	3950.93
	1006	Advanced Robotics for Manufacturing	2100000.00	12.0	175000.00	1666.6667	2216.9600	3883.63
	1009	Historical Preservation Techniques	1100000.00	12.0	91666.67	1666.6667	2208.3000	3874.97
	1004	Next-Gen Quantum Computing	4200000.00	24.0	175000.00	833.3333	2961.7200	3795.05

16. Track project keyword trends over the past 5 years

SQL Command:

```

SELECT k.k_type AS keyword,
    YEAR(p.p_contract_start_date) AS project_year,
    COUNT(p.p_id) AS projects_using_keyword,
    SUM(p.budget_amount) AS total_budget_with_keyword

```

```

FROM Keyword k
JOIN Project_has_Keyword phk ON k.k_id = phk.Keyword_k_id
JOIN Project p ON phk.Project_p_id = p.p_id AND phk.Project_Donor_d_id = p.Donor_d_id
WHERE p.p_contract_start_date >= DATE_SUB(CURDATE(), INTERVAL 5 YEAR)
GROUP BY k.k_type, YEAR(p.p_contract_start_date)
ORDER BY keyword, project_year DESC;

```

```

274 • SELECT k.k_type AS keyword,
275         YEAR(p.p_contract_start_date) AS project_year,
276         COUNT(p.p_id) AS projects_using_keyword,
277         SUM(p.budget_amount) AS total_budget_with_keyword
278 FROM Keyword k
279 JOIN Project_has_Keyword phk ON k.k_id = phk.Keyword_k_id
280 JOIN Project p ON phk.Project_p_id = p.p_id AND phk.Project_Donor_d_id = p.Donor_d_id
281 WHERE p.p_contract_start_date >= DATE_SUB(CURDATE(), INTERVAL 5 YEAR)
282 GROUP BY k.k_type, YEAR(p.p_contract_start_date)
283 ORDER BY keyword, project_year DESC;
284
285
286
287

```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	keyword	project_year	projects_using_keyword	total_budget_with_keyword
▶	AI	2024	1	2500000.00
	algorithm	2024	1	4200000.00
	artifacts	2024	1	1100000.00
	automation	2024	1	2100000.00
	biodiversity	2024	1	1500000.00
	childhood	2024	1	1200000.00

17. For all ongoing projects, show the total number of installments, the number of incomplete installments, and the number of complete installments.

[SQL Command:](#)

```

SELECT p.p_id, p.p_title, COUNT(i.i_id) AS total_installments, SUM(CASE WHEN i.i_status = 'Complete' THEN 1 ELSE 0 END)
AS complete_installments, SUM(CASE WHEN i.i_status = 'Incomplete' THEN 1 ELSE 0 END) AS incomplete_installments
FROM Project p
LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
WHERE p.p_status = 'On going'
GROUP BY p.p_id, p.p_title
ORDER BY p.p_title;

```

```

415 • SELECT p.p_id,
416         p.p_title,
417         COUNT(i.i_id) AS total_installments,
418         SUM(CASE WHEN i.i_status = 'Complete' THEN 1 ELSE 0 END) AS complete_installments,
419         SUM(CASE WHEN i.i_status = 'Incomplete' THEN 1 ELSE 0 END) AS incomplete_installments
420 FROM Project p
421 LEFT JOIN Installment i ON p.p_id = i.Project_p_id AND p.Donor_d_id = i.Project_Donor_d_id
422 WHERE p.p_status = 'On going'
423 GROUP BY p.p_id, p.p_title
424 ORDER BY p.p_title;
425
426

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	p_id	p_title	total_installments	complete_installments	incomplete_installments
▶	1006	Advanced Robotics for Manufacturing	3	3	0
	1008	Cybersecurity Protocol Development	2	2	0
	1007	Early Childhood Education Models	2	2	0
	1010	Genomic Sequencing of Rare Diseases	2	2	0
	1009	Historical Preservation Techniques	2	2	0
	1011	Microbial Fuel Cells	3	3	0

Result 25 x

18. Identify projects that have passed their contract end date but are still 'On going'.

SQL Command:


```

SELECT p.p_id, p.p_title, p.p_contract_end_date, DATEDIFF(CURDATE(), p.p_contract_end_date) AS days_overdue, d.d_name
AS donor_name

FROM Project p

JOIN Donor d ON p.Donor_d_id = d.d_id

WHERE p.p_status = 'On going'

AND p.p_contract_end_date < CURDATE()

ORDER BY days_overdue DESC;

```

```

427 • SELECT p.p_id,
428         p.p_title,
429         p.p_contract_end_date,
430         DATEDIFF(CURDATE(), p.p_contract_end_date) AS days_overdue,
431         d.d_name AS donor_name
432 FROM Project p
433 JOIN Donor d ON p.Donor_d_id = d.d_id
434 WHERE p.p_status = 'On going'
435        AND p.p_contract_end_date < CURDATE()
436 ORDER BY days_overdue DESC;
437
438

```

result Grid   Filter Rows: Export:  Wrap Cell Content: 				
p_id	p_title	p_contract_end_date	days_overdue	donor_name
1006	Advanced Robotics for Manufacturing	2025-08-15 17:00:00	21	National Science Fund
1002	Renewable Energy Grid Optimization	2025-09-01 17:00:00	4	Government Innovation Agency

19. For all ongoing projects, calculate the total estimated budget to be spent in each year (based on the contract end date) [SQL Command:](#)

```
SELECT YEAR(p.p_contract_end_date) AS forecast_year, COUNT(p.p_id) AS projects_ending, SUM(p.budget_amount) AS total_budget
```

```
FROM Project p
```

```
WHERE p.p_status = 'On going'
```

```
GROUP BY YEAR(p.p_contract_end_date)
```

```
ORDER BY forecast_year;
```

```
441 • SELECT YEAR(p.p_contract_end_date) AS forecast_year,  
442         COUNT(p.p_id) AS projects_ending,  
443         SUM(p.budget_amount) AS total_budget  
444 FROM Project p  
445 WHERE p.p_status = 'On going'  
446 GROUP BY YEAR(p.p_contract_end_date)  
447 ORDER BY forecast_year;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	forecast_year	projects_ending	total_budget
▶	2025	4	11500000.00
	2026	5	13200000.00
	2027	1	3000000.00

20. For each ongoing project, show the project lead (Principal Investigator), the total number of staff, and the project budget, to help in planning staff for new projects with similar budgets.

SQL Command:

```
SELECT p.p_id, p.p_title, p.budget_amount, CONCAT(s.s_firstname, ' ', s.s_lastname) AS principal_investigator,
staff_count.total_staff

FROM Project p

LEFT JOIN Staff s ON p.p_id = s.Project_p_id AND p.Donor_d_id = s.Project_Donor_d_id AND s.s_role = 'Principle Investigator'

LEFT JOIN (

    SELECT Project_p_id, Project_Donor_d_id, COUNT(*) AS total_staff

    FROM Staff

    GROUP BY Project_p_id, Project_Donor_d_id

) staff_count ON p.p_id = staff_count.Project_p_id AND p.Donor_d_id = staff_count.Project_Donor_d_id

WHERE p.p_status = 'On going'

ORDER BY p.budget_amount DESC;
```

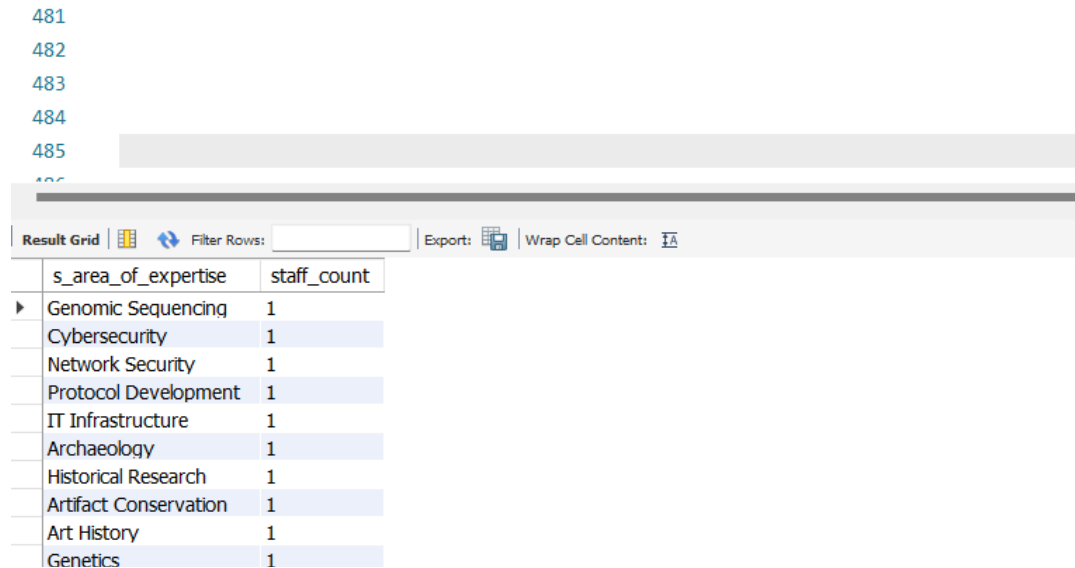
p_id	p_title	budget_amount	principal_investigator	total_staff
1002	Renewable Energy Grid Optimization	3500000.00	David Lee	4
1003	Sustainable Urban Farming	1800000.00	Maria Garcia	4
1004	Next-Gen Quantum Computing	4200000.00	John Smith	4
1006	Advanced Robotics for Manufacturing	2100000.00	Michael Brown	4
1007	Early Childhood Education Models	1200000.00	Jessica Wilson	4
1008	Cybersecurity Protocol Development	4800000.00	Daniel Miller	4
1009	Historical Preservation Techniques	1100000.00	Olivia Moore	4
1010	Genomic Sequencing of Rare Diseases	3200000.00	Laura Evans	4
1011	Microbial Fuel Cells	2800000.00	Robert Clark	4
1012	Smart City Traffic Management	3000000.00	Megan Hall	4

21. Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

SQL Command:

```
SELECT s.s_area_of_expertise, COUNT(s.s_id) AS staff_count
FROM Staff s
JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
WHERE p.p_status = 'On going'
GROUP BY s.s_area_of_expertise
ORDER BY staff_count ASC;
```

```
474 • SELECT s.s_area_of_expertise,
475         COUNT(s.s_id) AS staff_count
476 FROM Staff s
477 JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
478 WHERE p.p_status = 'On going'
479 GROUP BY s.s_area_of_expertise
480 ORDER BY staff_count ASC;
```



The screenshot shows a database query result grid with the following data:

s_area_of_expertise	staff_count
Genomic Sequencing	1
Cybersecurity	1
Network Security	1
Protocol Development	1
IT Infrastructure	1
Archaeology	1
Historical Research	1
Artifact Conservation	1
Art History	1
Genetics	1

22. Identify available staff members (not currently part of an ongoing project) who can be assigned to new projects.

SQL Command:

```
SELECT DISTINCT s.s_id, CONCAT(s.s_firstname, ' ', s.s_lastname) AS staff_name, s.s_area_of_expertise, d.dp_name AS department
```

```
FROM Staff s
```

```
JOIN Department d ON s.Department_dp_id = d.dp_id
```

```
JOIN Project p ON s.Project_p_id = p.p_id AND s.Project_Donor_d_id = p.Donor_d_id
```

```
WHERE p.p_status = 'Complete'
```

```
AND s.s_id NOT IN (
```

```
    SELECT s2.s_id
```

```
    FROM Staff s2
```

```
    JOIN Project p2 ON s2.Project_p_id = p2.p_id AND s2.Project_Donor_d_id = p2.Donor_d_id
```

```
    WHERE p2.p_status = 'On going'
```

```
)
```

```
ORDER BY s.s_area_of_expertise, staff_name;
```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	s_id	staff_name	s_area_of_expertise	department
▶	3	David Lee	Algorithm Development	Engineering
	2	Laura Evans	Data Science	Research
	19	Olivia Moore	Ecosystem Modeling	Administration
	1	Alex Chen	Machine Learning	Research
	17	Emily Davis	Marine Biology	Administration
	20	Rebecca Thomas	Marine Conservation	Sciences
	10	Nancy Shaffer	Geography	Sciences

